

SESSION 3

Hardware & Software + Programming Languages

Parent Study Guide — Help Your Child Learn at Home

Ages 6 – 10 | Online Class

Dear Parents,

This guide helps you review Session 3 with your child. We explored the difference between Hardware (the things you can touch) and Software (the programs inside). We also introduced Programming Languages and how we use them to talk to computers!

What's Inside This Guide

1

Hardware

The parts you can touch and see

2

Software

The instructions and programs inside

3

The Rule

An easy way to tell them apart

4

Programming Languages

How we talk to the computer

5

Activities & Homework

Practice games for home

1. Hardware

HARDWARE = The Body

Hardware is any part of the computer that you can TOUCH, HOLD, or DROP.

Ø=PK Analogy: If a computer is like a person, Hardware is the body (bones, eyes, hands).

Examples of Hardware

Monitor: The screen you look at.

Keyboard & Mouse: The tools you use to click and type.

CPU (Processor): The brain chip inside the computer.

Headphones: The device you wear to listen to sound.

2. Software

SOFTWARE = The Brain & Thoughts

Software is the programs, apps, and instructions inside the computer. You CANNOT touch it physically.

Ø>Ýà Analogy: If a computer is like a person, Software is the thoughts, memories, and ideas inside the brain.

Examples of Software

Games: Roblox, Minecraft, or puzzle games.

Web Browsers: Google Chrome or Safari.

Apps: YouTube, Zoom, or a drawing app like MS Paint.

Operating System: Windows, macOS, or iOS that runs the device.

The Golden Rule:

Hardware needs Software to know what to do.

Software needs Hardware to run.

They work as a TEAM!

3. Programming Languages

How do we talk to the computer?

As humans, we communicate with each other using human languages like English, Arabic, Spanish, or French.

But computers don't understand human language. To talk to a computer and tell it what to do, we use a Programming Language!

Human Languages

Used to talk to other humans.

Examples:

- English
- Arabic
- French

Programming Languages

Used to talk to computers.

Examples:

- Python
- JavaScript
- Scratch

👉 When we write instructions for the computer using a Programming Language, we are creating Software!

Summary of How It Works:

1. We learn a Programming Language.
2. We use it to write instructions (Code/Software).
3. The computer's Hardware runs the instructions.
4. The computer does what we asked!

4. Practice & Homework

Activity: Hardware or Software?

Read each item with your child. Ask them: "Can you touch it?" Then decide if it is Hardware or Software.

Item	Answer
Mouse	HARDWARE
YouTube	SOFTWARE
Keyboard	HARDWARE
Minecraft Game	SOFTWARE
Monitor	HARDWARE
Zoom App	SOFTWARE

Homework Task

Ask your child to find 3 pieces of Hardware in your house, and name 3 pieces of Software (like apps on a phone or tablet). Have them explain why they are Hardware or Software.

Coming Up Next...

In the next session, we will start exploring our first Programming Language and write our first code!